Bachelor Of Engineering In Information Technology

2^{nd} Year 2^{nd} Semester, First Class Test, 2026

Subject Name –Graph Theory & Combinatorics (IT/PC/B/T/224)

Time=1 Hours Full Marks=30

CO1 [10]

Q1.(i) Define a path in a graph. Explain how to obtain the path matrix of length 2 from the adjacency matrix of the graph shown in Figure 1.

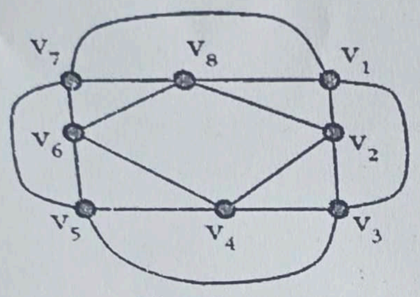


Figure 1

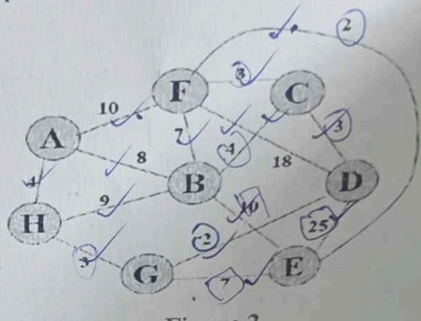


Figure 2

(ii) A graph has six vertices, and their degrees are: 3, 3, 2, 2, 2, 2

a) Determine whether the graph contains an Euler path or Euler circuit.

b) Justify your answer.

[5 +5=10]

CO2 [10]

Q2. (i) Prove "G is a tree if and only if there is a unique path between any two vertices."

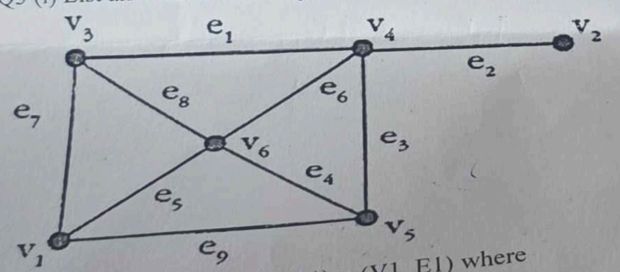
(ii) Use Prim's algorithm to find the minimal spanning tree of the graph shown in Figure 2.

(iii) Find the rank, nullity, and centre of the graph shown in Figure 2.

[3+4+3=10]

CO3 [10]

Q3 (i) List all cut sets with respect to two vertices V_1 and V_2.



e2

4

(ii) Given two graphs: Graph(G1) = (V1, E1) where

V1={A,B,C,D,E},E1={AB,AC,AD,BC,BD,CE,DE}

Graph(G2)=(V2,E2), V2={A,B,C,D,E}, E2={AB,AC,AE,BC,CD,DE,BE}

(a) Find the union(G1∪G2). List all edges of the resulting graph and draw it.

(b) Find the ring sum(G1⊕G2). List all edges of the resulting graph and draw it.

(iii) Justify this using a suitable example "The vertex connectivity of any graph G can never exceed the edge connectivity of G".

[4+4+2=10]

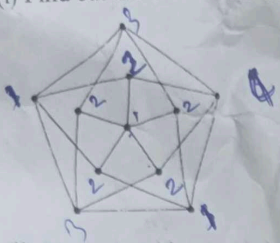
discuss the concept of different types of Graphs with fundamental properties

Bachelor Of Engineering In Information Technology

2nd Year 2nd Semester, 2nd Class Test, 2026

Subject Name –Graph Theory & Combinatorics (IT/PC/B/T/224)

Time=1 Hours **Full Marks=30**

CO4 [10]	Q1. (i) Find out the chromatic number of the graph given below (Mention all steps properly).  (ii) Show that if a graph has chromatic number k then it contains a subgraph with minimum degree at least k-1. (iii) Prove that every perfect matching is a maximum matching. [4+3+3=10]
CO5 [10]	**Q2.** (i) A drawer contains a dozen brown socks and a dozen black socks, all unmatched. A man takes socks out at random in the dark. How many socks must he take out to be sure that he has at least two socks of the same color? (ii) How many bit strings contain exactly eight 0s and ten 1s if every 0 must be immediately followed by a 1? (iii) How many different strings can be made from the letters in ABRACADABRA, using all the letters? (iv) Suppose there are 21 faculty members in the CSE department and 24 in the IT department. How many ways are there to select a committee consisting of 6 CSE faculty and 4 IT faculty? [2+2+3+3=10]
CO6 [10]	**Q3. (i)** Find the generating functions of the following sequences in closed form. $\langle 2, -4, 6, -8, 10, -12, \ldots\ldots\ldots \rangle$ (ii) Find the sequence generated by the following generating functions: $$\frac{1}{1-4x}$$ (iii) Using generating functions, find a_n in terms of n for the case given below: $$a_0 = 1,\ a_1 = 2 \text{ and } a_{n+2} = 5a_{n+1} - 4a_n \text{ for } n \geq 0$$ [4+4+2=10]

After completing this course the students should be able to:

CO4: Illustrate planar graph and their properti...